

Global Geopolitical Event Prediction: Analyzing the Impact on the USD/IDR Exchange Rate



Christiano Jose Intoro - 24/546891/PA/23217

Panji Merah Balakosa - 24/532826/PA/22541

Aditya Evan Rahmadani - 24/536755/PA/22771

Github:

Reason for choosing the news source

We had 2 realistic options on the approved list **GDEL**T and **CNBC** and we actually tried both before deciding, instead of picking one on a hunch. CNBC gave us one newsroom's take on the world, while GDEL

T covers over 12,000 outlets globally and tags every article with a fixed topic code instead of a keyword, so "geopolitical" means the same thing in 2021 as it does in 2026. The deciding factor, though, was practical, which we timed both. GDEL

T pulled 2,020 files in about 13 minutes with zero errors. CNBC's live search only got through 20 of 30 search terms in 28+ minutes, and a full run was projected at roughly 8 hours, about 37x slower, with a real chance of getting blocked partway through since CNBC's robots.txt explicitly disallows automated finance crawlers. Given we need to re-run this pipeline multiple times across the next three tasks, GDEL

T was the only option that didn't put the whole project at risk.

Scraping Technique

This actually covers two different acquisitions, since we have two streams:

- **News:** GDEL
- T doesn't need "scraping" in the traditional sense, it publishes pre-built bulk files, so we just download the file GDEL
- T already made rather than rendering pages or running live searches. That's the whole reason it's fast: no search, no pagination, just a direct file grab, run in parallel.
- **Exchange rate:** here we hit a real snag. The plan was to hit bi.go.id's JISDOR page directly, month by month, and parse the HTML table. In practice, BI's site resets the connection the moment it sees a non-browser client, that's a deliberate bot defense, not a bug in our code. So instead we pull the same interbank spot series from a source that mirrors that exact market, and we keep a small sample of real BI values on the side to check the two line up. We didn't quietly swap sources and hope nobody noticed, it's written down, and there's a reconciliation script that reports the gap.

Filtering strategy

Raw GDEL

T is enormous and mostly irrelevant, so we run everything through three filters, cheapest first:

- **Topic filter:** keep only articles tagged with a theme code we care about (armed conflict, sanctions, monetary policy, elections, trade, energy/commodities). Codes instead of keywords, because keyword vocabulary drifts over five years and a fixed code doesn't.
- **Relevance filter:** even a "geopolitical" article isn't automatically about USD/IDR, so we require a real connection: US/the Fed (the dollar side), Indonesia/BI (the rupiah side), China/ASEAN (Indonesia's main trade partners), and commodities Indonesia actually exports (coal, palm oil, nickel).
- **Duplicate collapse:** GDEL
- T indexes the same wire story hundreds of times as it gets syndicated. We collapse those into one record, but we don't throw away the copy

count. a story that ran 200 times is more newsworthy than one that ran twice, so that number becomes a usable "salience" signal instead of getting deleted as noise.

Every layer logs how many articles went in and came out, so if a later result looks weak, we can tell whether that's the model's fault or an overly aggressive filter.

Data cleaning and preprocessing strategy

- Some cleaning is shared no matter what, drop anything with a broken date, missing title, or missing URL. get every timestamp onto one timezone before doing anything else, strip HTML junk, cookie banners, and "related articles" clutter, fix broken character encoding, and flag (not delete) non-English articles.
- After that point, the text splits into two cleaned versions, because simple statistical models and transformer models want opposite things from the same sentence. statistical models want the fat trimmed, lowercase everything, strip punctuation, drop stopwords, reduce words to their root form, replace numbers with a placeholder, because those extra tokens just add noise for a bag-of-words model. Transformer models wants the opposite, because that "noise" is exactly what a transformer was trained to read: capitalization flags names, punctuation marks where a sentence ends, and stopwords like "not" carry the actual meaning ("no deal was reached" completely flips if you strip "no"). So we keep both versions side by side instead of forcing one compromise.
- On the numbers side, parse dates with an explicit format (never let the computer guess), fix Indonesia's period-as-thousands-separator formatting so "1.234" doesn't get misread as one-point-two-three-four, keep the last corrected value if BI revises a rate, and flag suspicious jumps for a human to look at rather than silently deleting them. And we never invent a rate for a day the market was simply closed, for example a missing Sunday isn't missing data.

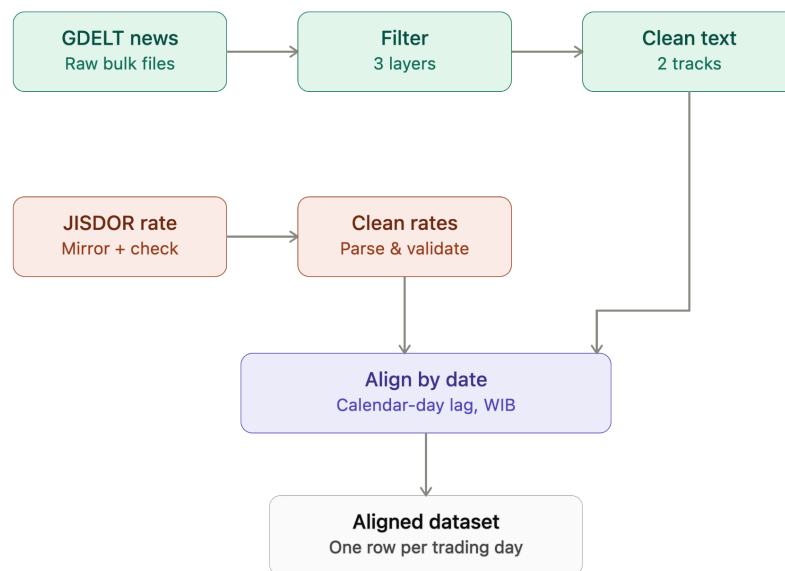
Alignment rules for timezones and non-trading days

- News flows continuously throughout the day, but JISDOR is only published once per business day. Worse, the publication time itself hasn't been stable. Bank Indonesia has changed the announcement time three separate times since 2021. This means there's no single, consistent intraday cutoff we could use across the whole historical dataset. Trying to reconstruct and apply the exact cutoff time for each historical regime would require us to defend three different sets of assumptions, and any mistake in pinning down those exact times would introduce errors that are hard to catch and easy to argue with.
- Instead of chasing exact cutoff times, we used a simpler day-level rule, whatever WIB calendar day a news item was published on, it gets matched with the next trading day's rate. If that next day happens to fall on a weekend or holiday, the match rolls forward to the next actual trading day, so all the news that piled up during the closed period gets bucketed together with that same next available rate. The one technical

wrinkle that mattered here is that GDELT timestamps come in UTC, not WIB. We add 7 hours to convert to WIB before extracting the calendar day. Doing the conversion in the other order (extracting the day first, then adjusting) would misfile a meaningful chunk of US-market news, since US news often lands late in the UTC day and would get assigned to the wrong WIB calendar day if converted incorrectly.

- This rule sidesteps the need to track which of BI's three publication-time regimes applied at any given point in history. Then, it produces correct bucketing automatically no matter which regime was in effect, because it never depends on knowing the exact intraday cutoff. Just as important, it guarantees no lookahead bias, news can never be paired with a rate that was already set before that news existed. Eventhough it's a coarser, day-level rule rather than a precise minute-level one, it trades a bit of precision for something more defensible and robust across the entire dataset.

Visual overview



Stage	Output	Recorded
Raw GDELT records	All indexed articles in window	Row count per file, per day
After Layer 1 (topic)	Geopolitically themed articles	Count + retention rate per theme code
After Layer 2 (relevance)	Articles with a USD/IDR channel	Count + retention rate per entity

After Layer 3 (dedup)	Unique stories with copy_count	Unique story count + duplication ratio
-----------------------	-----------------------------------	---